Digital Business

Topic 2

Digital Business Infrastructure & Digital Payments

Learning Objectives

2.1 Discuss the origins of, and the key technology concepts behind, the Internet.

2.2 Explain the current structure of the Internet.

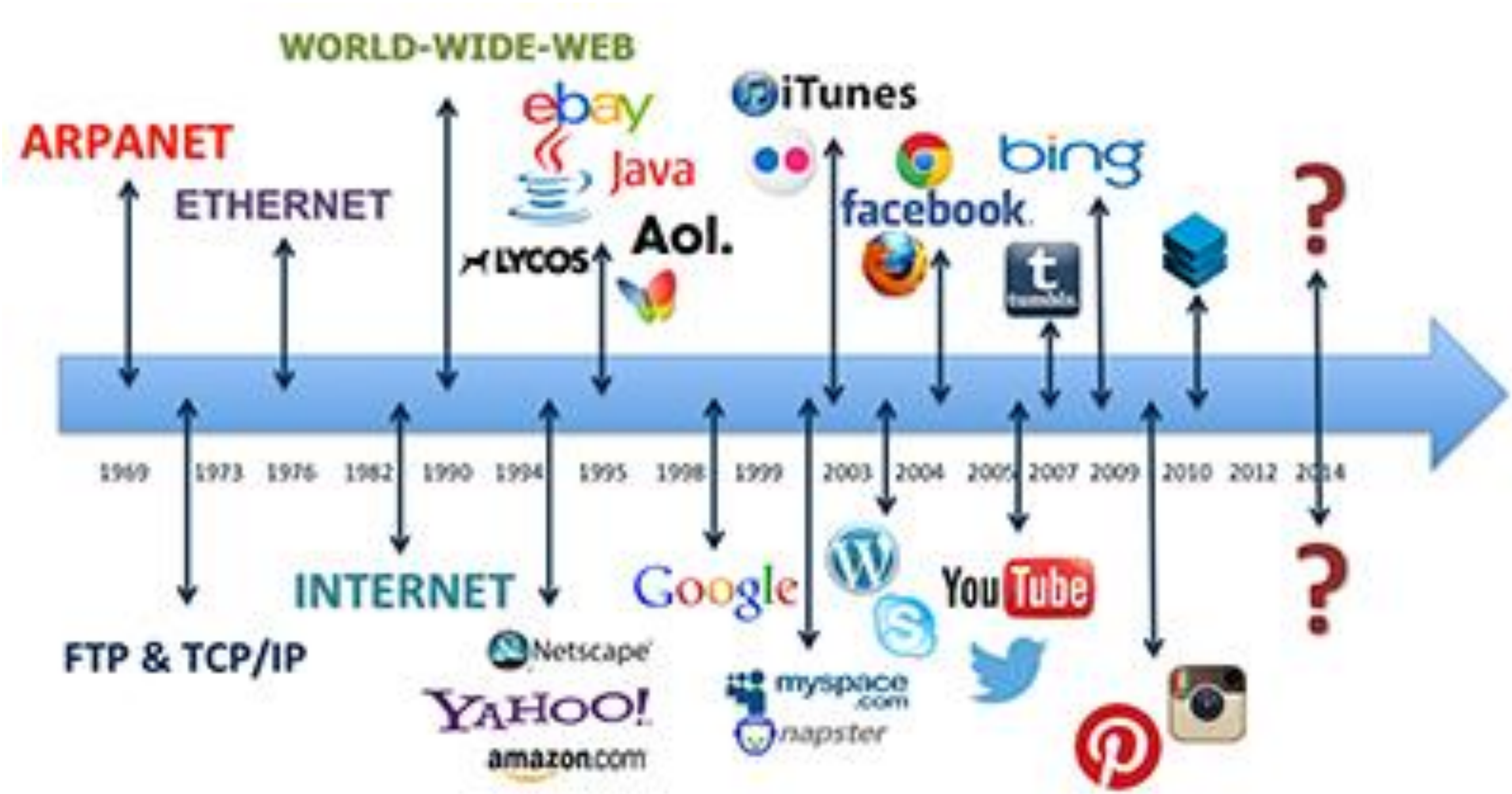
2.3 Understand how the Web works.

2.4 Describe how Internet and web features and services support e-business.

The Internet: Technology Background

- Internet
  - Interconnected network of thousands of networks and millions of computers
  - Links businesses, educational institutions, government agencies, and individuals
- World Wide Web (Web)
  - One of the Internet's most popular services
  - Provides access to billions, possibly trillions, of web pages

The Evolution of the Internet 1961–Present



https://malonemediagroup.com/history-of-the-internet-timeline-an-ever-evolving-digital-world/


HISTORY
OF THE INTERNET

The Internet: Key Technology Concepts

- Internet defined as network that:
  - Uses IP addressing
  - Supports TCP/IP
  - Provides services to users, in manner similar to telephone system
- Three important concepts:
  - Packet switching
  - TCP/IP communications protocol
  - Client/server computing

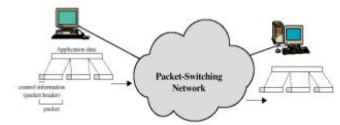
TCP/IP

- Transmission Control Protocol (TCP)
  - Establishes connections among sending and receiving Web computers
  - Handles assembly of packets at point of transmission, and reassembly at receiving end
- Internet Protocol (IP)

Packet Switching

- Slices digital messages into packets
- Sends packets along different communication paths as they become available
- Reassembles packets once they arrive at destination
- Uses routers
- Less expensive, wasteful than circuit-switching

Packet Switching

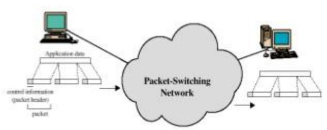


Around 1970, research began on a new form of architecture for long distance communications: Packet Switching.

Figure 2.3 Packet Switching

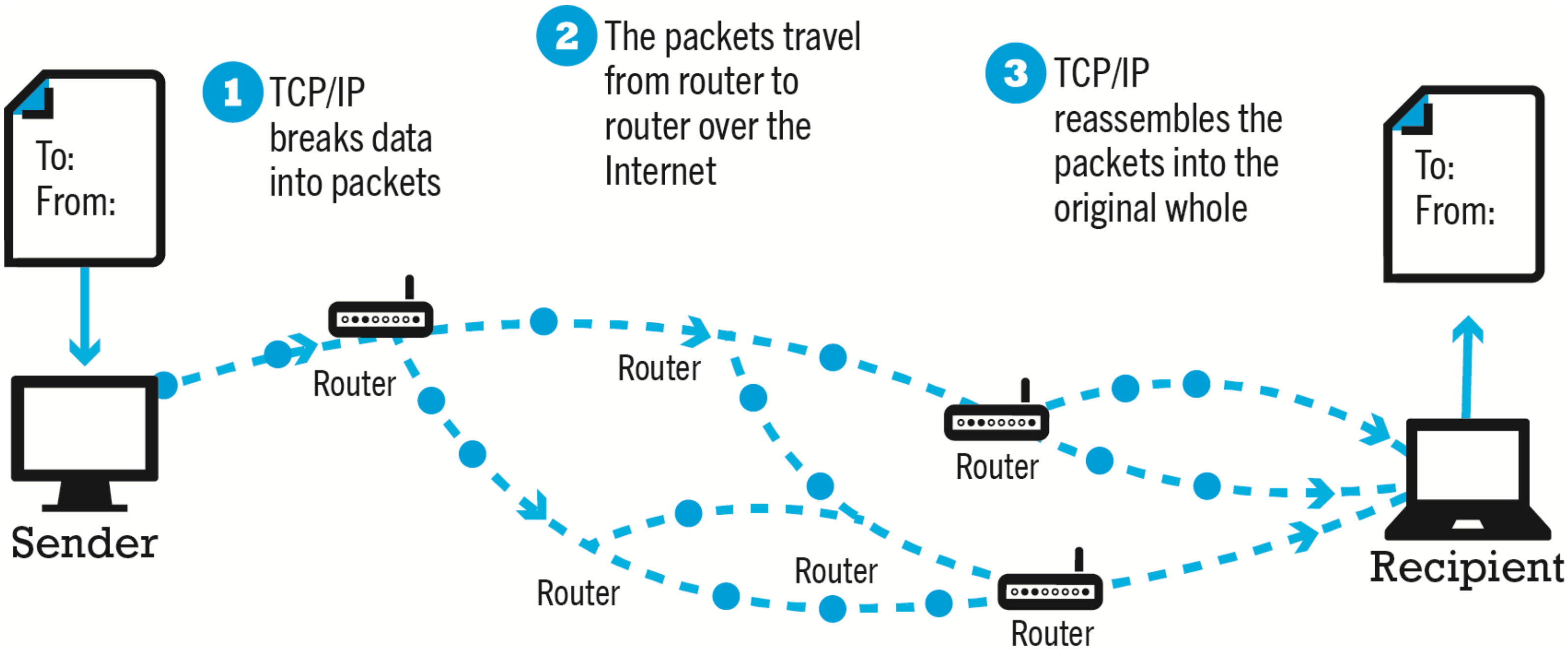
I want to communicate with you.	Original text message
1011000100110111 0001101	Text message digitized into bits
10110001 00110111 0001101	Digital bits broken into packets
0011001 10110001 00110111 0001101	Header information added to each packet indicating destination and other control information, such as how many bits are in the total message and how many packets

Packet Switching



Around 1970, research began on a new form of architecture for long distance communications: Packet Switching.

Figure 2.5 Routing Internet Messages: TCP/IP and Packet Switching



Internet (IP) Addresses

- IPv4
  - 32-bit number
  - Four sets of numbers marked off by periods: 201.61.186.227
    - Class C address: Network identified by first three sets, computer identified by last set
- IPv6
  - 128-bit addresses, able to handle up to 1 quadrillion addresses (IPv4 can handle only 4 billion)

Domain Names, DNS, and URLs

- Domain name
  - IP address expressed in natural language
- Domain name system (DNS)
  - Allows numeric IP addresses to be expressed in natural language
- Uniform resource locator (URL)
  - Address used by Web browser to identify location of content on the Web
  - For example: http://www.amazon.com

Connectivity

Client/Server Computing

- Powerful personal computers (clients) connected in network with one or more servers
- Servers perform common functions for the clients
  - Storing files
  - Software applications
  - Access to printers, and so on

Wireless Local Area Network (WLAN) - Based Internet Access

- Wi-Fi (various 802.11 standards)
  - High-speed, fixed broadband wireless LAN (WLAN)
  - Wireless access point (“hot spots”)
  - Limited range but inexpensive
- WiMax
- Bluetooth

Figure 2.13 Wi-Fi Networks

https://techviral.net/allow-only-selected-device-to-connect-on-wifi/

Mobile Internet Access

- Two basic types of wireless Internet access:
  - Telephone-based (mobile phones, smartphones)
  - Computer network-based (wireless local area network-based)
- Telephone-based wireless Internet access
  - Currently based on 3G and 4G technologies
  - 5G will provide higher bandwidth with speeds reaching 10 Gbps or more

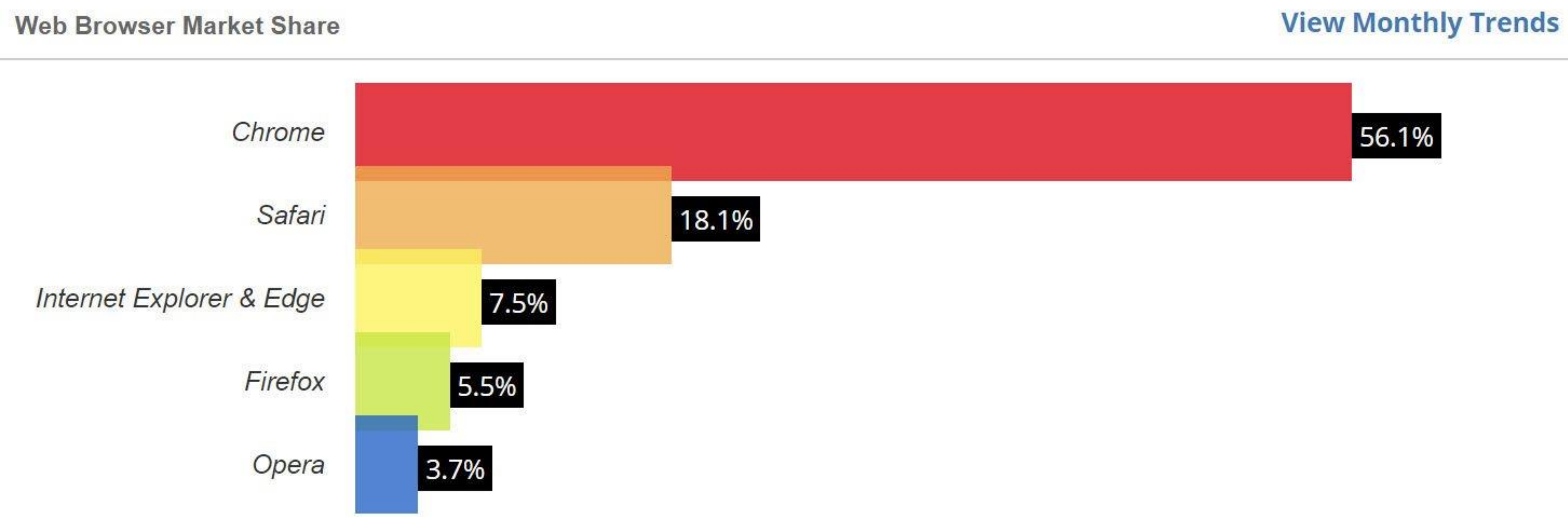
The Internet and Web: Features

The Internet and Web: Features

- Features on which the foundations of e-business are built:
  - Web browsers
  - Mobile platform
  - Cloud Computing
  - IoT
  - Communication tools
  - Search engines
  - Downloadable and streaming media
  - Web 2.0 applications and services
  - Virtual reality and augmented reality
  - Intelligent digital assistants

Web Browsers

Web Browser Market Share



https://www.techadvisor.co.uk/test-centre/software/best-web-browsers-3635255/

The Mobile Platform

- Primary Internet access is now through tablets and smartphones
- Tablets supplement PCs for mobile situations
  - Over 160 million people in U.S. use Internet with tablets
- Smartphones are a disruptive technology
  - New processors and operating systems
  - Over 230 million in U.S. access Internet with smartphones

Mobile Apps

- Use of mobile apps has exploded
  - Most popular entertainment media, over TV
  - Always present shopping tool
  - Almost all top 100 brands have an app
- Platforms
  - iPhone/iPad (iOS), Android
- App marketplaces
  - Google Play, Apple's App Store, Amazon's Appstore

The Internet "Cloud Computing" Model (1 of 2)

- Firms and individuals obtain computing power and software over Internet
- Three types of services
  - Infrastructure as a service (IaaS)
  - Software as a service (SaaS)
  - Platform as a service (PaaS)
- Public, private, and hybrid clouds

The Internet "Cloud Computing" Model (2 of 2)

- Drawbacks
  - Security risks
  - Shifts responsibility for storage and control to providers
- Radically reduces costs of:
  - Building and operating websites
  - Infrastructure, IT support
  - Hardware, software

The Internet of Things (IOT)

- Objects connected via sensors/RFID to the Internet
- “Smart things”
- Interoperability issues and standards
- Security and privacy concerns

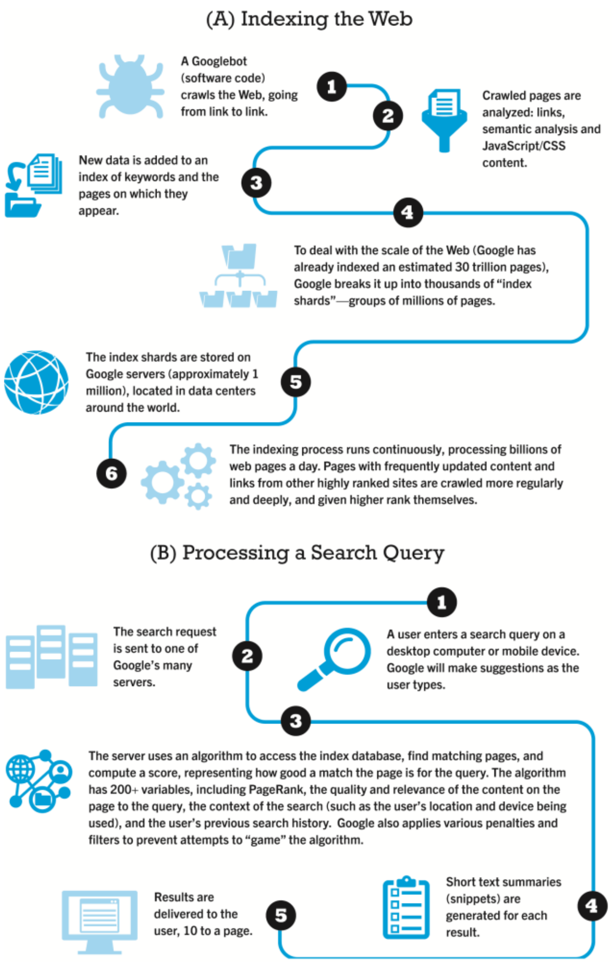
Communication Tools

- E-mail
  - Most used application of the Internet
- Messaging Applications
  - Instant messaging
- Online message boards
- Internet telephony
  - VOIP
- Video conferencing, video chatting, telepresence

Search Engines

- Identify web pages that match queries based on one or more techniques
  - Keyword indexes
  - Page ranking
- Also serve as:
  - Shopping tools
  - Advertising vehicles (search engine marketing)
  - Tool within e-commerce sites
- Top three providers: Google, Bing, Yahoo (Oath)

Figure 2.17 How Google Works



Downloadable and Streaming Media

- Downloads:
  - Growth in broadband connections enables large media file downloads
- Streaming technologies
  - Enables music, video, and other large files to be sent to users in chunks so that the file can play uninterrupted
- Podcasting
- Explosion in online video viewing

Web 2.0 Features and Services

- Online Social Networks
  - Services that support communication among networks of friends, peers (Facebook, Tik Tok, Instagram)
- Blogs
  - Personal web page of chronological entries
  - Enables web page publishing with no knowledge of HTML
- Wikis
  - Enables documents to be written collectively and collaboratively
  - E.g. Wikipedia

Virtual Reality and Augmented Reality

- Virtual reality (VR)
  - Immersing users within virtual world
  - Typically uses head-mounted display (HMD)
  - Oculus Rift, Vive, PlayStation VR
- Augmented reality (AR)
  - Overlaying virtual objects over the real world, via mobile devices or HMDs
  - Pokémon GO

Intelligent Digital Assistants

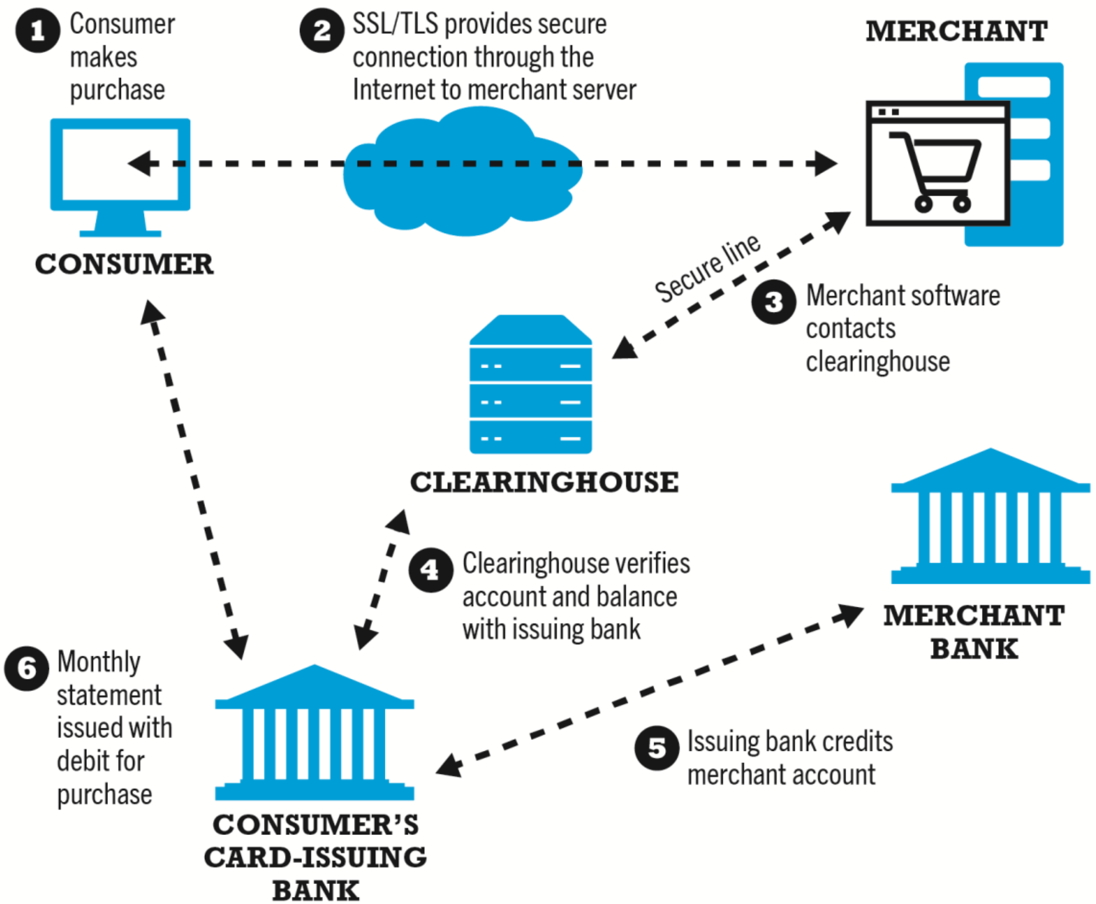
- Computer search engine using:
  - Natural language
  - Conversational interface, verbal commands
  - Situational awareness
- Can handle requests for appointments, flights, routes, event scheduling, and more.
  - Examples:
    - Apple's Siri
    - Google Now
    - Google Assistant

Digital Payment Systems

E-commerce Payment Systems

- In U.S., credit and debit cards are primary online payment methods
  - Other countries have different systems
- Online credit card purchasing cycle
- Credit card e-commerce enablers
- Limitations of online credit card payment
  - Security, merchant risk
  - Cost
  - Social equity

Figure 4.14 How an Online Credit Card Transaction Works



Alternative Online Payment Systems

- Online stored value systems:
  - Based on value stored in a consumer’s bank, checking, or credit card account
  - Example: PayPal
- Other alternatives:
  - Amazon Pay
  - Visa Checkout, Mastercard’s MasterPass
  - Dwolla

Mobile Payment Systems

- Use of mobile phones as payment devices
  - Established in Europe and Asia
  - Expanding in United States
- Near field communication (NFC)
- Different types of mobile wallets
  - Universal proximity mobile wallets, such as Apple Pay, Google Pay, Samsung Pay, PayPal Mobile
  - Branded store proximity wallets, offered by Walmart, Target, Starbucks, others
  - P2P mobile payment apps, such as Zelle, Venmo

Cryptocurrencies

- Use blockchain technology and cryptography to create a purely digital medium of exchange
- Bitcoin the most prominent example
  - Value of Bitcoins have widely fluctuated
  - Major issues with theft and fraud
  - Some governments have banned Bitcoin, although it is gaining acceptance in the U.S.
- Other cryptocurrencies (altcoins) include Ethereum/Ether, Ripple, Litecoin and Monero
- Initial coin offerings (ICOs) being used by some startups to raise capital

Questions?